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Implement your Own Ping Program

Aim:

To implement a ping program that checks the reachability of a destination host and measures its response time.

Algorithm/procedure:

1. Select a programming language and import the required networking libraries.
2. Obtain the destination hostname or IP address from the user.
3. Resolve the hostname to an IP address if necessary
4. Create an ICMP Echo Request Packet with an Identifier, Sequence number, and timestamp.
5. Send the request to the Destination host using an appropriate ICMP - Capable Socket.
6. wait for an ICMP Echo Reply or until a timeout occurs.
7. Calculate the round-trip time using the request and reply timestamps
8. Display the destination IP address, response time, and packet Status.

Sample Input:

Enter IP address: 8.8.8.8

Sample output:

Pinging 8.8.8.8

Reply received
Reply received
Reply received
Reply received

Packets Sent: 4
Packets received: 4
Packet Loss: 0%.

a. Repeat the process for a specified number of requests

b. Calculate packet loss and summarize the results.

Result:

A ping program was implemented successfully to test host reachability, measure round trip time, and identify packet loss.